AWS for Data Engineering
Absolute Beginners

From Cloud Concepts to Real Tools

CONTINUATION SESSION

Building upon: **Cloud Computing Fundamentals**

Including Architecture, Security, and Data Roles

Amazon Web Services

Data Engineering

The "Problem-First" Approach

We won't just memorize tool names. Instead, we'll start with a real-world problem, understand the general cloud concept that solves it, and then reveal the specific AWS tool.

CRITICAL TEACHING PRINCIPLE

AWS services are just answers to common problems. If you understand the problem, the tool makes sense automatically.

1. The Problem

e.g., "Where do we put all these messy files?"

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2. The Cloud Concept

e.g., "A Data Lake (Digital Warehouse)"

↓

3. The AWS Service

e.g., "Amazon S3"

SMOOTH TRANSITION

From Cloud Computing to AWS

We learned the **concepts** (WHAT).
Now we learn the **platform** (WHERE).

Concept → aws Provider

Why AWS for Data Engineering?

The Problem

Data is growing too fast for traditional servers to handle.

- **Volume:** Too much data
- **Velocity:** Coming too fast
- **Variety:** Messy formats

Cloud Concept

We need unlimited, on-demand power without buying hardware.

- **Scalability:** Grow instantly
- **Pay-as-you-go:** Rent, don't buy
- **Managed:** No maintenance

INDUSTRY STANDARD

AWS Solution

The world's most widely used cloud platform for data.

- **S3:** Unlimited storage
- **EC2:** Resizable compute
- **Glue & Redshift:** ETL & Analytics

Think of it like moving houses:

Instead of buying a bigger house every time you get more stuff (Traditional), you rent a magic storage unit that automatically expands when you put boxes in it (AWS Cloud).

The AWS Big Picture

5 Key Pillars for Data Engineering

1. Storage

Where we keep the raw data files safely and cheaply.

aws Amazon S3

2. Compute

The raw processing power (virtual machines) to run tasks.

aws Amazon EC2

3. Processing (ETL)

Cleaning, sorting, and preparing data for use.

aws AWS Glue

4. Analytics

Asking questions and getting answers from data.

aws Amazon Redshift

5. Machine Learning

Teaching computers to predict future trends.

aws Amazon SageMaker

THE KITCHEN ANALOGY

Think of a professional restaurant: **Storage (S3)** is the pantry, **Compute (EC2)** is the electricity/gas, **Processing (Glue)** is the prep chefs chopping veggies, **Analytics (Redshift)** is the head chef organizing plates, and **ML (SageMaker)** is creating new recipes.

Where Do We Store Everything?

Imagine a giant warehouse. You have millions of items—raw materials, boxes, photos, logs—and you don't want to organize them perfectly yet. You just need a cheap, safe place to dump them until they are needed.

> **THE CLOUD CONCEPT MAP**
>
> **The Concept:** A "Data Lake" (Digital Warehouse).
> **The Service:** Amazon S3 (Simple Storage Service). It stores any amount of data, anywhere, forever.

The Problem

"We have terabytes of raw, messy files (CSVs, Images, Logs) and no place to put them."

↓

The Concept: Data Lake

A "Digital Warehouse". Store raw data now, analyze it later. No structure required.

↓

The Service: Amazon S3

Unlimited storage buckets. Highly durable (won't lose data) and very cheap.

The Need for Processing Power

Big Data is too heavy for a regular laptop. Trying to process millions of records on your personal computer would make it freeze or crash. We need a way to get "super-computer" power just for the time we need it.

THE ANALOGY: A RENTED FACTORY

Think of EC2 as **renting a digital factory**. Instead of building your own factory (buying expensive servers), you rent machines by the hour to do the heavy lifting, then turn them off when the job is done.

1. The Problem

"My laptop crashes when opening a 1TB file."

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2. The Concept: Virtual Machine

A "computer inside a computer" that lives in the cloud.

↓

3. The Service: Amazon EC2

Elastic Compute Cloud – Rentable computing power.

AWS IAM – Access & Security

Who is knocking at the door?

- Controls **who** can enter the cloud account (Authentication).
- Controls **what** they are allowed to do (Authorization).
- Works like a building security system with ID badges and keycards.

LAYMAN ANALOGY

Think of IAM as the **Security Guard & Key Master**. Just because you can enter the building (Login) doesn't mean you have the key to the CEO's office (Permissions).

1. The Problem

"We can't give everyone the master keys to the entire company."

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2. The Concept

Identity & Permissions (Users, Roles, Policies)

↓

3. The AWS Service

AWS IAM (Identity and Access Management)

AWS Glue – ETL Concepts

The Problem

Raw data is messy and cannot be used directly for analytics.

- **Messy Formats:** JSON, CSV mixed
- **Bad Data:** Missing values & errors
- **Duplicates:** Same user twice

ETL Concept

We need a process to clean and prepare data automatically.

- **Extract:** Pull from source
- **Transform:** Clean & change format
- **Load:** Save to warehouse

SERVERLESS ETL

AWS Glue

A serverless service that prepares data for analytics.

- **Automated:** Writes code for you
- **Cleans:** Fixes messy data
- **Scheduled:** Runs every night

Think of it like a Restaurant Kitchen:

Raw Ingredients (Data) come from the farm dirty.
The Kitchen (Glue) washes, chops, and cooks them (Transform) so they are ready to be served on a Plate (Data Warehouse)

Amazon Redshift – Data Warehousing

Data Lake

Amazon S3

ANALOGY: THE STOREROOM

Throw everything in boxes.
Cheap & Easy to store, but **Slow** to find specific things later.

- **Raw Data:** Images, JSON, CSV
- **Cost:** Very Low
- **Query Speed:** Slow for analytics

VS

Data Warehouse

Amazon Redshift

ANALOGY: THE LIBRARY

Books are cataloged on shelves.
Organized & Structured so you can find answers **Instantly**.

- **Structured Data:** Rows & Columns
- **Query Speed:** Extremely Fast
- **Purpose:** Analytics & Reports

Key Concept: We keep ALL data in the Lake (S3), but we move only the IMPORTANT data to the Warehouse (Redshift) for analysis.

Intro to SageMaker – ML on AWS

The Problem

Building smart AI is hard. It requires massive computer power and complex setups.

- **Hardware:** Need expensive GPUs
- **Complexity:** Hard to manage code
- **Deploying:** Hard to use models

ML Concepts

Separate the "learning" from the "doing" using scalable cloud resources.

- **Training:** Teaching the model
- **Inference:** Using the model
- **Pipeline:** Automating steps

AWS MACHINE LEARNING

Amazon SageMaker

A fully managed service to build, train, and deploy ML models quickly.

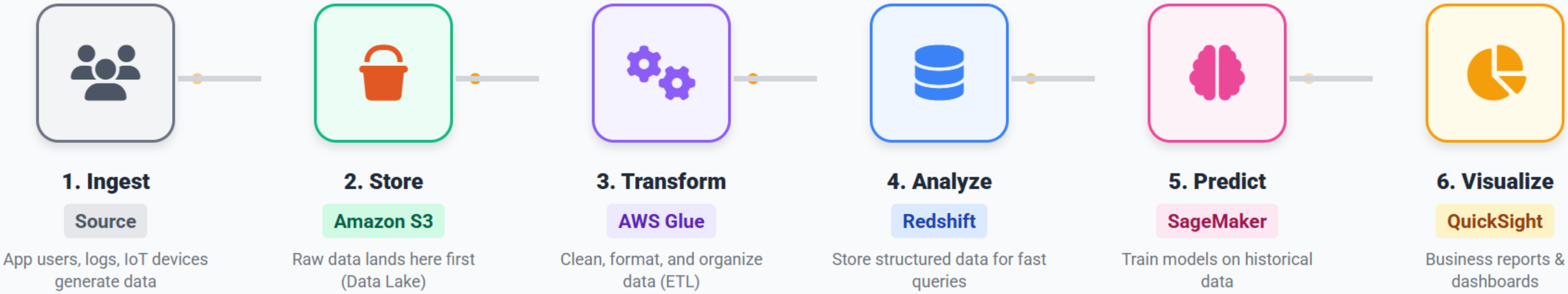
- **Build:** Ready-made notebooks
- **Train:** Rent supercomputers
- **Deploy:** Click to launch

Think of it like an "AI Factory":

You don't build the factory yourself. You just bring your blueprints (data & code), and SageMaker's assembly line automatically builds (trains) and packages (deploys) your robot for you.

End-to-End AWS Data Pipeline

A Typical Modern Data Workflow



1. **Raw Data** enters the system messy and unstructured.
2. **Glue** acts as the "Kitchen" to clean and prep ingredients.
3. **Redshift** serves the cooked meal to analysts instantly.

Who Does What? Mapping Roles to AWS

Connecting People to the Pipeline

DATA ENGINEER

The Builder

MAIN RESPONSIBILITY

Builds and maintains the "pipes" that move data. Ensures data flows reliably from source to destination.

AWS TOOLBOX

- S3 (Storage)
- Glue (ETL)
- EC2 (Compute)
- Redshift (Admin)

The Plumber
"I install the pipes so water (data) flows to the kitchen without leaking."

DATA ANALYST

The Investigator

MAIN RESPONSIBILITY

Uses structured data to answer business questions. Creates dashboards and reports for managers.

AWS TOOLBOX

- Redshift (Query)
- QuickSight
- Athena
- Excel / SQL

The Food Critic
"I taste the meal (data) and write a review explaining if it's good or bad."

DATA SCIENTIST

The Predictor

MAIN RESPONSIBILITY

Uses math and algorithms to find hidden patterns and predict future trends.

AWS TOOLBOX

- SageMaker
- S3 (Raw Data)
- Notebooks
- Python/R

The Fortune Teller
"I look at the past to tell you what will likely happen in the future."